

Report - End-to-End Machine Learning Pipeline Using Logistic Regression

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1. Executive Summary

This report presents an end-to-end machine learning pipeline developed to predict whether a business account is likely to upgrade its account. The project follows a structured workflow that includes business understanding, data loading, data inspection, data cleaning, missing value handling, feature and target separation, train-test splitting, preprocessing, Logistic Regression model training, prediction, evaluation, and business interpretation.

The dataset contains 377 original records and 13 columns. After removing 7 duplicate rows, the cleaned dataset contains 370 records. The target variable is `Upgraded_Account`, with 288 non-upgraded accounts and 82 upgraded accounts after cleaning.

The Logistic Regression baseline model achieved a testing accuracy of 75.68%. For the positive class, Yes, the model achieved precision of 41.67%, recall of 31.25%, and F1-score of 35.71%. The confusion matrix and classification report show that the model performs better for the No class than for the Yes class, which is likely connected to class imbalance in the target variable.

Overall, the model provides a useful starting point for identifying business accounts that may be suitable for sales follow-up, customer success support, training, engagement, or upgrade-related actions. However, because it is a baseline model and has weaker performance for the upgrade class, it should be used as a decision-support tool rather than as the only basis for business decisions.

2. Introduction and Assignment Objective

The objective of this assignment is to build a basic end-to-end machine learning pipeline using a provided business dataset. The project is not limited to training a model; it also demonstrates the full workflow required to move from raw data to an evaluated machine learning model.

The notebook and this report focus on a business account upgrade prediction problem. The purpose of the model is to predict whether a business account is likely to upgrade based on available account, usage, revenue, support, sales, and training information.

The model selected for this assignment is Logistic Regression. Logistic Regression is appropriate for this task because the target variable has two possible outcomes: Yes and No. Therefore, the problem is a supervised binary classification task.

3. Business Problem

Many companies offer different account plans or service levels. Some business accounts remain on their current plan, while others upgrade to a higher-value plan. Understanding which accounts are more likely to upgrade can help the company improve sales planning, customer relationship management, customer success activities, training support, and targeted marketing campaigns.

The key business question is:

Can the company predict whether a business account will upgrade using account information, revenue details, transaction activity, product usage, support history, sales contact information, and customer engagement indicators?

A model that identifies likely upgrade accounts can help the sales or customer success team prioritize outreach. For example, accounts predicted as more likely to upgrade may be prioritized for product demonstrations, account reviews, training support, or upgrade offers. Accounts predicted as less likely to upgrade may require different engagement strategies.

4. Dataset Overview

The dataset used for this project is `business_account_upgrade_prediction_dataset.csv`. Each row represents one business account. The dataset includes account-level information such as company size, industry, annual revenue, monthly transactions, current plan, account age, support tickets, sales contact status, product usage score, monthly fee, training attendance, and upgrade outcome.

Table 1: Dataset and project summary

Item	Value
Original dataset shape	377 rows x 13 columns
Duplicate rows before cleaning	7
Dataset shape after removing duplicates	370 rows x 13 columns
Final cleaned dataset shape	370 rows x 13 columns
Target variable	Upgraded Account
Problem type	Supervised binary classification
Model used	Logistic Regression baseline model

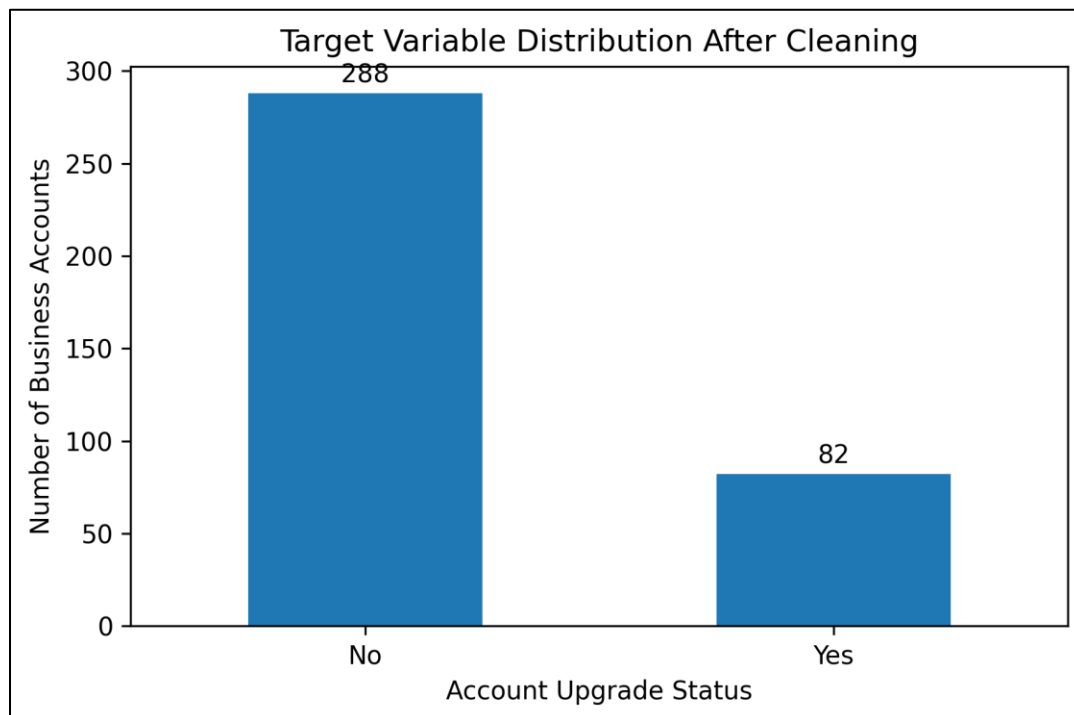


Figure 1: Target variable distribution after cleaning

Figure 1 shows that the cleaned dataset contains 288 accounts in the No class and 82 accounts in the Yes class. The No class is larger than the Yes class, which is important when interpreting model performance.

5. Target Variable and Input Features

The target variable for this assignment is `Upgraded_Account`. It shows whether a business account upgraded its account or did not upgrade. The possible values are Yes and No.

Table 2: Dataset columns and business meaning

Column	Description
Account_ID	Identifier only; removed before model training
Company_Size	Business size category
Industry	Industry category of the business account
Annual_Revenue	Estimated annual revenue of the account
Monthly_Transactions	Number of monthly transactions
Current_Plan	Current account plan or service level
Account_Age_Months	Length of account relationship in months
Support_Tickets	Number of support tickets
Sales_Contacted	Whether the account was contacted by sales
Product_Usage_Score	Product engagement or usage score
Monthly_Fee	Monthly fee paid by the account
Training_Attended	Whether the account attended training
Upgraded_Account	Target variable; Yes or No upgrade outcome

For model training, `Account_ID` was removed because it is only an identification number and does not provide meaningful predictive information. `Upgraded_Account` was separated as the target variable, while all remaining business variables were used as input features.

6. Data Preparation and Inspection

The dataset was inspected before model training to understand its structure and identify data quality issues. The inspection included reviewing the number of rows and columns, the first and last records, column names, data types, missing values, duplicate rows, numerical summary statistics, categorical summary statistics, and target class distribution.

Table 3: Numerical and categorical feature groups

Feature Type	Number of Features	Columns
Numerical	6	Annual_Revenue, Monthly_Transactions, Account_Age_Months, Support_Tickets, Product_Usage_Score, Monthly_Fee
Categorical	5	Company_Size, Industry, Current_Plan, Sales_Contacted, Training_Attended

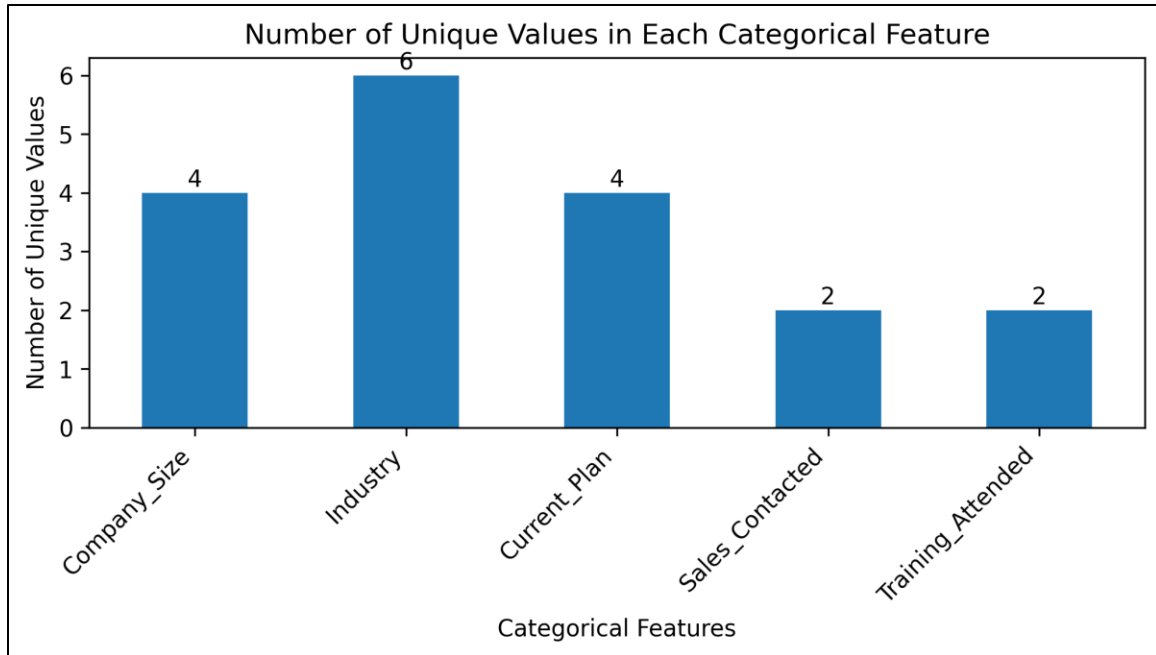


Figure 2: Number of unique values in each categorical feature

The data contains both numerical and categorical features. This requires different preprocessing methods: numerical features are scaled, while categorical features are converted into numerical format using one-hot encoding.

7. Data Cleaning and Missing Value Handling

The dataset contained 7 duplicate rows. These rows were removed because repeated records can give extra weight to the same observations and may affect model training. After duplicate removal, the dataset contained 370 records.

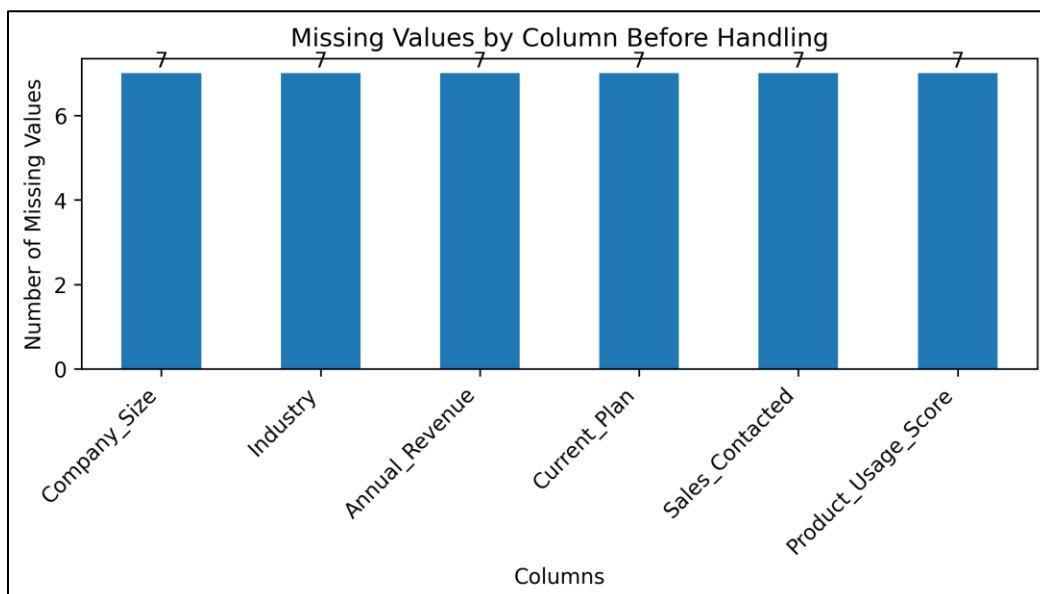


Figure 3: Missing values by column before handling

Missing values were handled after duplicate removal. The target variable did not contain missing values. Numerical missing values were filled using the median, while categorical missing values were filled using the mode. This approach allowed the dataset to retain useful records instead of removing rows unnecessarily.

Table 4: Missing value handling summary

Column	Data Type	Missing Values	Method	Fill Value
Annual Revenue	Numerical	7	Median	4,168,551.00
Product Usage Score	Numerical	7	Median	51.00
Company Size	Categorical	7	Mode	Small
Industry	Categorical	7	Mode	Technology
Current Plan	Categorical	7	Mode	Basic
Sales Contacted	Categorical	7	Mode	No

After missing value handling, the dataset contained no remaining missing values. The final cleaned dataset used for modeling contained 370 records and 13 columns.

8. Feature and Target Separation

After cleaning, the dataset was separated into X and y. X contains the input features used by the model, while y contains the target variable, Upgraded_Account. Account_ID was removed from X because it is only an identifier.

Table 5: Feature matrix and target vector

Object	Shape	Meaning
X	370 rows x 11 feature columns	Input variables used by the model
y	370 values	Target variable: Upgraded_Account

9. Train-Test Split and Preprocessing

The cleaned dataset was split into training and testing sets using an 80/20 split. The random_state parameter was set to 42 to make the split reproducible. The stratify=y option was used to keep the Yes and No class distribution similar in both training and testing sets.

Table 6: Train-test split summary

Dataset	Shape
X_train	(296, 11)
X_test	(74, 11)
y_train	(296,)
y_test	(74,)

Table 7: Target class distribution in training and testing sets

Target Class	Training Set	Testing Set
No	230	58
Yes	66	16

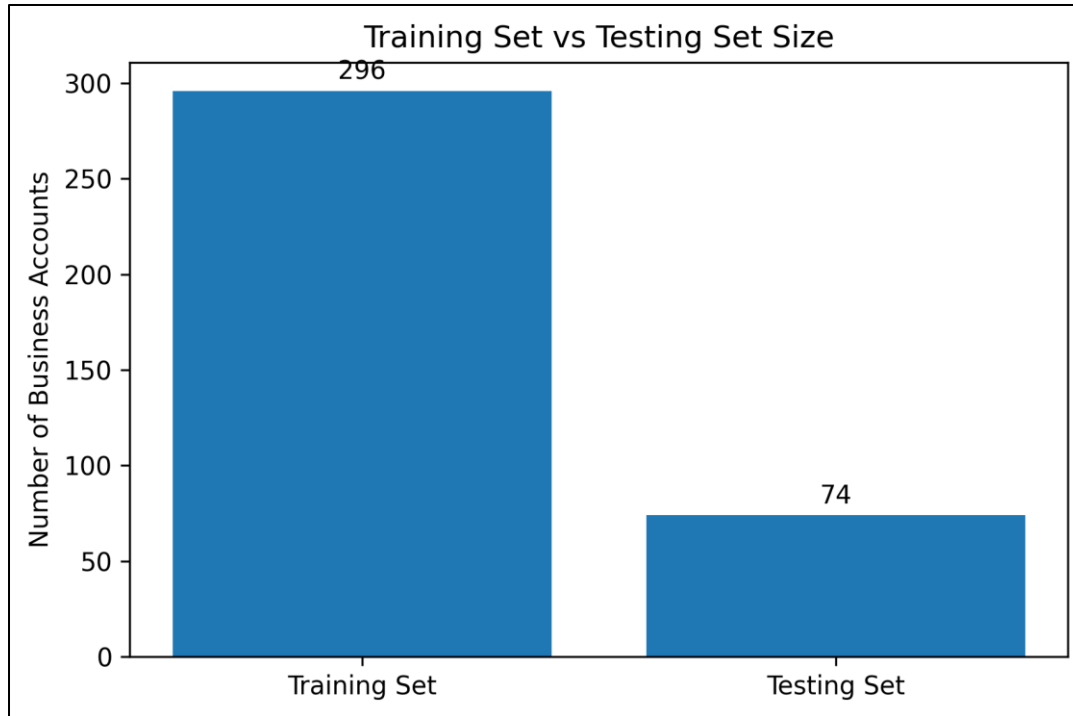


Figure 4: Training set vs testing set size

Preprocessing was applied using a ColumnTransformer. Numerical features were scaled using StandardScaler, and categorical features were encoded using OneHotEncoder with handle_unknown="ignore". This created a consistent preprocessing workflow for the Logistic Regression model.

Table 8: Preprocessing approach

Feature Group	Columns	Transformation
Numerical features	Annual_Revenue, Monthly_Transactions, Account_Age_Months, Support_Tickets, Product_Usage_Score, Monthly_Fee	StandardScaler
Categorical features	Company_Size, Industry, Current_Plan, Sales_Contacted, Training_Attended	OneHotEncoder

10. Logistic Regression Model Training and Prediction

A Logistic Regression model was trained as the baseline classification model. Logistic Regression was selected because it is simple, commonly used, and suitable for binary classification problems. In this project, the model predicts whether each business account is likely to upgrade.

The Logistic Regression model was placed inside a scikit-learn Pipeline together with the preprocessing steps. This keeps the workflow organized by applying preprocessing and model training in one structured process.

Table 9: Sample actual and predicted values

Actual	Predicted
No	No
No	No
Yes	No
No	No
No	No
No	No
Yes	Yes
No	No
Yes	Yes
No	No

11. Model Evaluation Results

The trained model was evaluated on the testing dataset. The predicted values were compared with the actual values using accuracy, precision, recall, F1-score, a classification report, and a confusion matrix.

Table 10: Logistic Regression evaluation metrics

Metric	Score	Percentage
Accuracy	0.7568	75.68%
Precision for Yes	0.4167	41.67%
Recall for Yes	0.3125	31.25%
F1-score for Yes	0.3571	35.71%

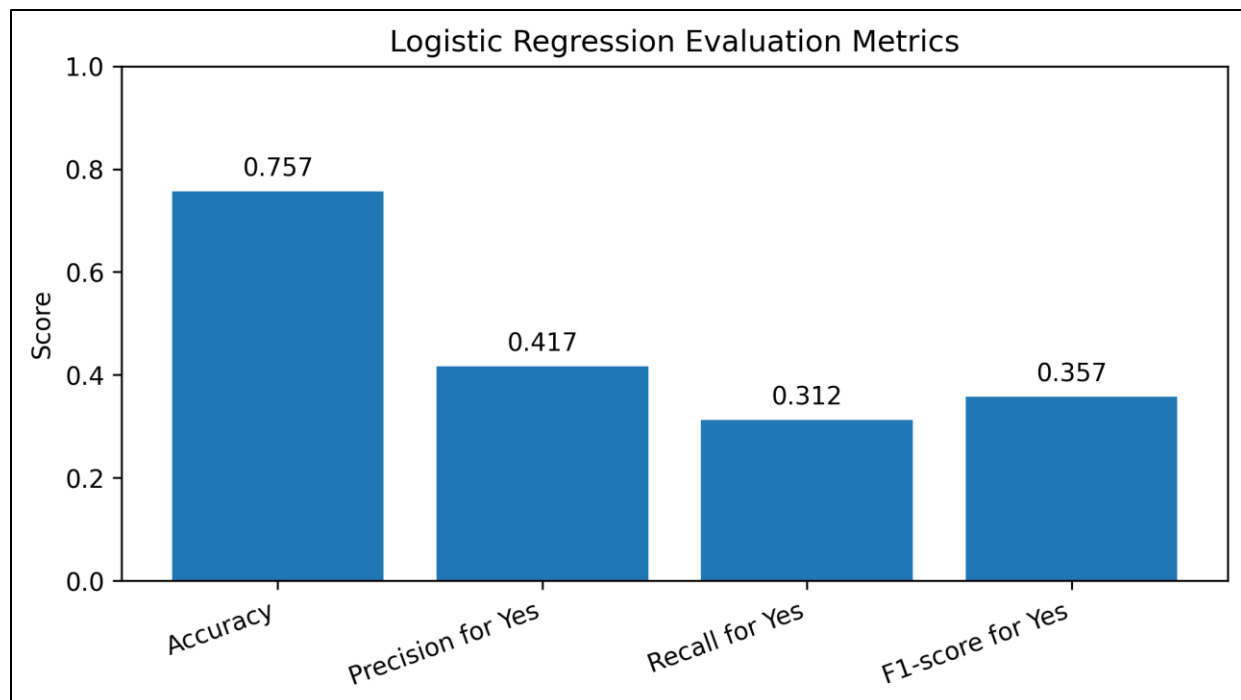


Figure 5: Logistic Regression evaluation metrics

Table 11: Classification report

Class/Metric	Precision	Recall	F1-score	Support
No	0.8226	0.8793	0.8500	58
Yes	0.4167	0.3125	0.3571	16
accuracy			0.7568	74
macro avg	0.6196	0.5959	0.6036	74
weighted avg	0.7348	0.7568	0.7434	74

Table 12: Confusion matrix values

Actual / Predicted	Predicted No	Predicted Yes
Actual No	51	7
Actual Yes	11	5

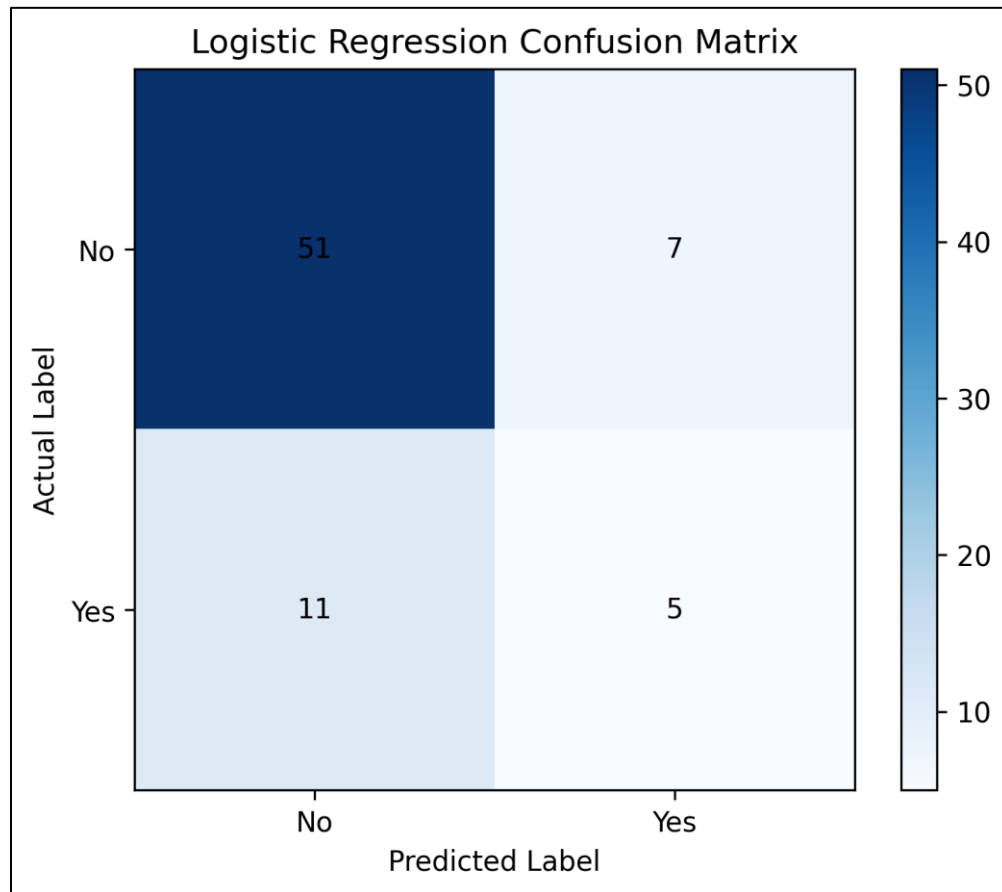


Figure 6: Logistic Regression confusion matrix

The confusion matrix shows that the model correctly predicted 51 non-upgraded accounts and 5 upgraded accounts. It incorrectly predicted 7 non-upgraded accounts as upgraded and 11 upgraded accounts as not upgraded. This indicates that the model is stronger at identifying the No class than the Yes class.

12. Business Interpretation of the Results

The model achieved a testing accuracy of 75.68%, meaning it correctly classified about 76% of the testing records. However, accuracy alone does not fully explain model performance because the dataset has more non-upgraded accounts than upgraded accounts.

The precision for the Yes class was 41.67%. This means that when the model predicted an account as upgraded, it was correct about 42% of the time. The recall for the Yes class was 31.25%, meaning the model identified about 31% of the actual upgraded accounts.

From a business perspective, accounts predicted as likely to upgrade may be considered for sales follow-up, account review, product demonstrations, training support, or targeted upgrade campaigns. However, because the recall for the Yes class is relatively low, the company should not rely on the model alone to identify all upgrade opportunities.

A false negative means the model predicts that an account is unlikely to upgrade, but the account actually belongs to the upgrade class. In business terms, this may cause the company to miss an account that could have been prioritized for sales follow-up or upgrade-related engagement. False positives also matter because the business may spend sales resources on accounts that are less likely to upgrade.

12.1 Training Accuracy, Testing Accuracy, and Generalization

In addition to testing performance, the model's training accuracy and testing accuracy were compared to understand how well the model generalizes to unseen data. Training accuracy shows how well the model performs on the data it learned from, while testing accuracy shows how well it performs on new records that were not used during training.

The Logistic Regression model achieved a training accuracy of 89.53% and a testing accuracy of 75.68%. The difference between the two values is approximately 13.85%. This suggests that the model performs better on the training data than on the testing data.

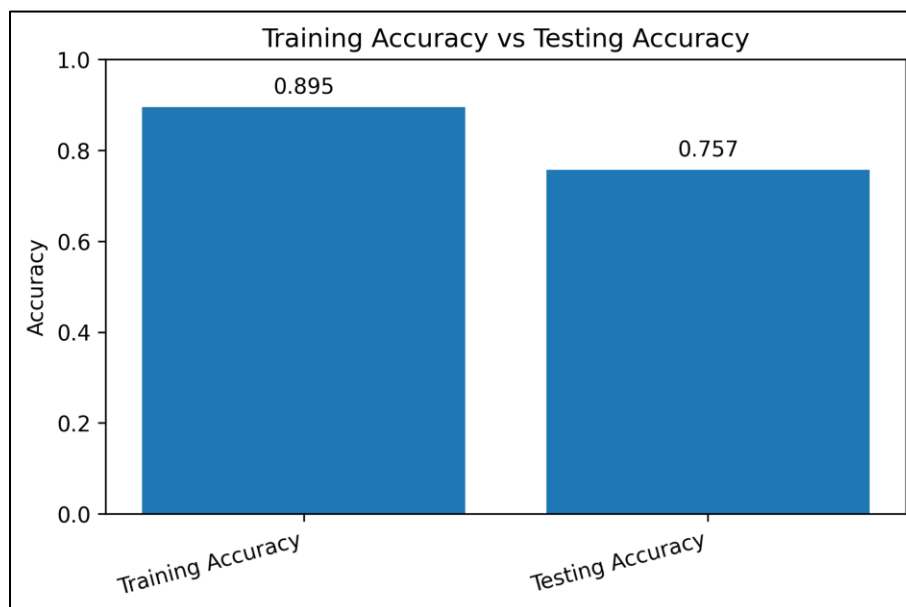


Figure 7: Training accuracy vs testing accuracy

Table 13: Training and testing accuracy comparison

Evaluation Set	Accuracy	Percentage
Training Set	0.8953	89.53%
Testing Set	0.7568	75.68%
Difference	0.1385	13.85%

The training-testing accuracy gap suggests that the model may have learned some patterns from the training data that do not fully generalize to unseen business accounts. Since Logistic Regression is used here as a baseline model, this result should be interpreted as an initial performance benchmark rather than a final production-ready model. Future work could use cross-validation, class balancing methods, or compare additional classification models to assess generalization more carefully.

13. Model Limitations

The first limitation is class imbalance. The cleaned dataset contains more No values than Yes values. As a result, the model performs better for the majority class and is weaker at identifying accounts that actually upgraded.

The second limitation is that the model is based only on the variables available in the dataset. Real upgrade decisions may depend on additional factors that are not included, such as customer budget, competitor offers, contract negotiations, satisfaction with service quality, leadership decisions, or future expansion plans.

The third limitation is that Logistic Regression is a simple baseline model. While it is useful for establishing a first performance level, it may not capture more complex relationships among variables. Future work could compare Logistic Regression with Decision Tree, Random Forest, Support Vector Machine, Gradient Boosting, or other classification models.

A further methodological improvement would be to place missing value imputation inside the modeling pipeline and fit it only on the training data. This would reduce the risk of data leakage in future versions of the project.

14. Responsible AI Reflection

This model should be used as a decision-support tool rather than as the only basis for business decisions. The model can help identify patterns, but it cannot fully understand customer context, business relationships, contract negotiations, or strategic account considerations.

The company should not use the model to unfairly ignore, exclude, or deprioritize customers. A business account predicted as unlikely to upgrade may still become an upgrade opportunity later. Similarly, a business account predicted as likely to upgrade may not upgrade due to reasons not captured in the dataset.

Responsible use requires human review, continuous performance monitoring, bias checking, and careful interpretation of model outputs. If used in a real business environment, the model should be tested on newer data and evaluated regularly to ensure that predictions remain fair, accurate, and useful.

15. GitHub Submission and Reproducibility

The GitHub repository includes the completed Jupyter Notebook, dataset, README file, generated output charts, and the report. The notebook is organized so that another user can run the full pipeline from data loading to model evaluation.

16. Final Conclusion

This project completed an end-to-end machine learning pipeline for business account upgrade prediction using a Logistic Regression baseline model. The workflow included business understanding, data loading, data inspection, data cleaning, missing value handling, feature and target separation, train-test split, preprocessing, model training, prediction, evaluation, result interpretation, limitation discussion, and responsible AI reflection.

The final Logistic Regression model achieved a testing accuracy of 75.68%. The evaluation results show that the model is useful as an initial baseline, but it performs better for the non-upgrade class than for the upgrade class. Therefore, the model can support business decision-making, but it should not replace human judgment.

Overall, the assignment demonstrates how a structured machine learning pipeline can transform business data into a practical prediction model that supports data-driven business decision-making.